

Ishak Tifouti

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Contents

1	Curriculum Vitae	2
1.1	Research interests	2
1.2	Research experience	2
1.3	Teaching experience	2
1.4	Education	2
1.5	Computing skills	3
1.6	Collective responsibilities	3
1.7	Languages	4
2	Scientific contributions	4
2.1	Publications	4
2.2	Software development	4
2.3	Conference presentations	4
2.4	Seminar and working group presentations	4
2.5	Workshops and scientific schools	5

1 Curriculum Vitae

Updated May 5th, 2026.

1.1 Research interests

My research lies at the interface between numerical analysis of partial differential equations and scientific computing. It revolves around model order reduction methods, error estimates-based mesh adaptation, adjoint methods and goal-oriented approximation, as well as data-driven approaches. During my PhD, I developed algorithms coupling these approaches — nonlinear model reduction via registration, parametric mesh adaptation and adjoint methods — for the construction of low-rank approximations of solutions to parametric compressible flows, modelled by the steady Euler equations.

In my postdoctoral work, I focus on wave propagation problems, particularly in the context of inverse problems, where I work on integrating reduced-order models into the inversion process for medium parameter identification, with a specific focus on full waveform inversion for seismic imaging.

1.2 Research experience

Postdoctoral researcher December 2025 – present

Inria Makutu Team — Université de Pau et des Pays de l'Adour

Topic: *Reduced-order models for inverse problems in seismic imaging*

Collaborators: Florian Faucher, Hélène Barucq, Ha Pham and Matteo Marcos

Doctoral researcher with teaching duties October 2022 – November 2025

Inria — Institut de Mathématiques de Bordeaux

Thesis: *Construction of reduced-order models with mesh adaptation*

Supervisors: Nicolas Barral and Tommaso Taddei

Master's research internship (M2) April 2021 – November 2021

CNAM Paris — Michelin R&D

Thesis: *Non-intrusive model reduction for a fluid-structure interaction problem*

Supervisors: Iraj Mortazavi, Florian De Vuyst and Thibault Dairay

1.3 Teaching experience

Qualification for the position of Maître de Conférences

Section 26 — Applied Mathematics and Applications of Mathematics

11/03/2026

Reviewer n°1: Ms Caroline Poisard

Reviewer n°2: Mr Mehdi Badsì

Tutorial sessions — Mathematical tools

L1 Mathematics, Université de Bordeaux

32 HETD — 2024

Tutorial sessions — Mathematical tools

L1 Mathematics, Université de Bordeaux

32 HETD — 2023

1.4 Education

PhD in applied mathematics October 2022 – November 2025

Inria Bordeaux — Institut de Mathématiques de Bordeaux

Thesis: *Construction of reduced-order models with mesh adaptation*

Supervisors: Nicolas Barral, Tommaso Taddei

Defense: November 20, 2025, Institut de Mathématiques de Bordeaux
PhD Committee:

Name	Position	Institution	Role
Anca-Claudia BELME	Associate Professor	Sorbonne Université	Reviewer
Olga MULA	Professor	University of Vienna	Reviewer
Frédéric ALAUZET	Research Director	INRIA	Examiner
Virginie EHRLACHER	Professor	École des Ponts ParisTech	Examiner
Luc MIEUSSENS	Professor	Bordeaux INP	President
Masayuki YANO	Associate Professor	University of Toronto	Examiner
Nicolas BARRAL	Associate Professor	Bordeaux INP	Supervisor
Tommaso TADDEI	Research Scientist	INRIA	Supervisor

Introduction to science communication

May 2023 — 7.5 h

Inria Center Training

Creation of outreach materials, science mediation for school students and the general public

Master's courses in Analysis, Modelling, Simulation

September 2021 – January 2022

Université Paris Saclay - ENSTA Paris — Institut Polytechnique de Paris

Specialisation: PDEs and Numerical Analysis

Master of Mathematical Modelling

2019 – 2021

Sorbonne Université, Paris

Specialisation: PDEs and Numerical Analysis

Bachelor of Mathematics

2016 – 2019

Sorbonne Université

1.5 Computing skills

- Python, MATLAB, Fortran, FreeFEM++
 - Scientific computing on remote servers (SSH, HPC clusters)
-

1.6 Collective responsibilities

- Member of the *Lambda* association for PhD students at the Institut de Mathématiques de Bordeaux
Responsible for organising doctoral seminars, September 2023 – January 2025
- European university network *ENLIGHT*
Participation in a stay at the University of Bern in May 2025 as part of the ENLIGHT network, aimed at strengthening partnerships between European universities, following a prize awarded by the Université de Bordeaux during the Quinzaine de l'Égalité.
- Hosting of College scholars in 2024 & 2025 for an introduction to public research careers, Inria Bordeaux
- Teaching Assistant — programme for students with disabilities
Sorbonne Université, Student Health Services
September 2018 – June 2021
- Library Assistant — Medical Sciences Library
Sorbonne Université
September 2018 – June 2021

1.7 Languages

French Bilingual
Arabic Bilingual
English C1 or above (Linguaskill Cambridge Certification)

2 Scientific contributions

2.1 Publications

Thesis

1. **I. Tifouti**. *Construction of reduced-order models with mesh adaptation*. PhD thesis, Institut de Mathématiques de Bordeaux, 2025.
HAL Id: <https://theses.hal.science/tel-05408853>

Journal article

1. N. Barral, T. Taddei, **I. Tifouti**. Registration-based model reduction of parameterized PDEs with spatio-parameter adaptivity.
Journal of Computational Physics, 2024.
<https://doi.org/10.1016/j.jcp.2023.112727>

Peer-reviewed conference proceedings

1. **I. Tifouti**, N. Barral, T. Taddei. Registration-based model reduction with local reduced order bases.
AIAA SciTech Forum, 2025.
<https://doi.org/10.2514/6.2025-1571>
HAL Id: <https://inria.hal.science/hal-05065511v1>

Papers in preparation

1. N. Barral, T. Taddei, **I. Tifouti**. Goal-oriented registration-based model reduction of shock-dominated flows with spatio-parameter adaptivity.
2. H. Barucq, F. Faucher, H. Pham, **I. Tifouti**. Reduced order modelling for wave propagation and inverse problems for full waveform inversion.

2.2 Software development

- Integration of model reduction methods into the open-source code **Hawen** (ongoing).
Code developed in Fortran by Florian Faucher, 2021.

2.3 Conference presentations

- ADMOS, Barcelona, 2025
- AIAA SciTech Conference, Orlando, 2025
- CANUM, La Rochelle, 2024

2.4 Seminar and working group presentations

- Journée Jeunes, Makutu Inria, Pau, 2026
- Memphis Seminar, Inria Bordeaux, La Rochelle, 2024
- Cardamom Seminar, Inria Bordeaux, Saint-Émilion, 2024
- Lambda PhD Seminar, Institut de Mathématiques de Bordeaux, 2023

2.5 Workshops and scientific schools

1. CEA–Inria–EDF Numerical Analysis Summer School, 2025
EDF Lab, Paris-Saclay.
2. Margaux PhD Days, May 2023
Poitiers.
3. MORTECH Workshop, November 2023
ENS Paris-Saclay.
4. ARIA Workshop on Industrial Applications of Model Order Reduction Techniques, March 2023
Inria Centre de l’Université de Bordeaux.